

Charlie Plumley

Independent Renewable Energy Consultant

Asset Performance | Data & Digitalisation | Technical Analytics | Applied AI

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Profile

Chartered Engineer, scientist and renewable energy specialist with 15 years’ technical experience spanning wind energy systems, operational asset performance, investment support, data strategy, automation and digitalisation. I help renewable energy investors, owners and operators turn complex technical and operational data into practical decisions, better workflows and improved asset performance.

My experience ranges from wind turbine controls and energy-yield analysis through to portfolio-wide performance analytics, SCADA-based monitoring, data architecture and the practical application of AI. Most recently, at Nuveen Infrastructure Clean Energy, I led data strategy and deployment of an in-house source-of-truth integrating SCADA, KPI, asset, financial and ESG data.

I combine detailed engineering understanding with investor- and owner-side experience, and work comfortably across technical, commercial and management teams.

How I Help Clients

Asset performance	Performance assessment, SCADA analysis, KPI development, loss analysis, benchmarking and identification of optimisation opportunities.
Data & digitalisation	Data strategy, renewable-energy data architecture, dashboards, automated reporting, analytics workflows and source-of-truth development.
Technical analytics	Energy-yield analysis and review, operational portfolio assessment, acquisition support, technical investigation and independent review.
AI & automation	Practical application of AI, coding and automation to renewable-energy workflows; training teams and developing internal capability.
Innovation & delivery	Technical problem-solving, R&D collaboration, project leadership and bridging between asset owners, investors, technology providers and academia.

Selected Track Record

- Led data strategy and deployment of an in-house source-of-truth for renewable investment funds, integrating SCADA, KPI, asset, financial and ESG data.
- Deployed dashboards and automated reporting to monitor and improve operational renewable asset performance.
- Monitored and improved renewable asset performance using SCADA data and additional instrumentation within an international owner/operator with a 3 GW operational portfolio across 10 countries.
- Conducted, supported and reviewed energy-yield assessments for acquisition and sale of assets, and brought operational energy-yield analysis in-house and automated the process.
- Championed open-source and open-data initiatives, including release of data from two wind farms and collaborations with industry and academia.
- Developed analysis to identify wind-farm optimisation opportunities and project-managed instrumentation of a 7 MW wind turbine.
- Designed and implemented a novel wind-turbine load-reduction control strategy to reduce gearbox wear.
- Trained colleagues on technology solutions and AI; presented at conferences and led external collaborations including battery revenue optimisation and power-prediction work.

Professional Experience

Glennmont | Nuveen Infrastructure Clean Energy **2024–2026**
Principal Digitalisation and Performance Engineer *London*

Renewable investment fund manager with portfolios of wind, solar and battery-storage projects.

- Managed data strategy and deployed an in-house source-of-truth covering SCADA, KPI, asset, financial and ESG data for the funds.
- Trained the team on technology solutions and AI.
- Presented at conferences and led external collaborations, including a battery revenue-optimisation project and power-prediction competition.

Glennmont | Nuveen Infrastructure Clean Energy **2023–2024**
Senior Performance Engineer *London*

- Deployed dashboards and automated reporting to monitor and improve performance of operational assets.
- Supported M&A work.

Cubico Sustainable Investments Ltd **2020–2023**
Performance Manager *London*

Owner, operator and investor in renewable assets; 3 GW operational portfolio across 10 countries.

- Monitored and improved operational asset performance using SCADA data and additional instrumentation.
- Championed open-source code and open-data projects, releasing data from two wind farms.
- Ran collaborations with industry and academia.

Cubico Sustainable Investments Ltd **2018–2020**
Energy Analyst *London*

- Conducted and reviewed energy-yield assessments for acquisitions.
- Brought energy-yield analysis for the operational portfolio in-house and automated the process.

SgurrControl | Wood PLC **2017–2018**
Control Algorithm Engineer *Glasgow*

- Developed analysis for identifying wind-farm optimisation opportunities.
- Project-managed instrumentation of a 7 MW wind turbine.

SgurrControl | Wood PLC **2015–2017**
Control Engineer *Glasgow*

- Designed and implemented a novel load-reduction control strategy to reduce wear on wind-turbine gearboxes.

Industry Leadership & Recognition

CEng mentor | WeDoWind founding member | 2025 NICE Spotlight Award | 2021 Cubico Diversity Award | Cranfield University CLEM IAB member | OpenOA contributor

Education & Professional Qualifications

2019–present	Chartered Engineer (CEng)
2014–present	Member, Institution of Engineering and Technology (IET)
2010–2015	PhD, Wind Energy Systems , University of Strathclyde — <i>Smart Rotor Control of Wind Turbines</i>
2009–2010	TORFL-1 Russian Language, Moscow State University
2005–2009	MPhys (Hons), Physics, First Class , University of Manchester

Technical Capability & Languages

Technical: Renewable asset performance, SCADA analysis, energy yield, wind-turbine controls, data analysis, reporting automation, coding, applied AI.

Professional: Project management, technical communication, collaboration, training and stakeholder engagement.

Languages: English; Russian; Spanish. (*Add proficiency descriptions if useful.*)